

American Calls and Puts under Black–Scholes–Merton from One-Driver Complex Geometric Brownian Motion

Phase-barrier geometry, a regime-aware boundary family, and a collocation pricing formula — Revision 2 after mathematical review

Complex–Itô Random-Walk Research Project

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Revision 2 of the Phase 15 paper, correcting every item of the external mathematical review of 2026-09-02. We price American calls and puts under Black–Scholes–Merton assumptions with continuous dividend yield q as European Black–Scholes plus the early-exercise premium (EEP; Kim, Jacka, Carr–Jarrow–Myneni) evaluated on a collocated boundary family. Revisions: (i) the put EEP is stated with its correct explicit sign (the one-line unified form needs an outer factor ϕ for puts); (ii) the short-expiry boundary asymptotics are the established three regimes of Evans–Kuske–Keller and Chen–Zhu — which our solved boundaries reproduce quantitatively — not a single universal law; (iii) the boundary family is redesigned to be dimensionless (declared logarithm scale $\hat{\tau}$), smooth on $(0, \infty)$, regime-aware, and monotone through the former artificial cusp at $\tau = 1$; (iv) all stationarity/envelope claims are removed — the substituted-boundary EEP functional is first-order sensitive to boundary error, demonstrated by perturbation; (v) every benchmark is recomputed against a converged Broadie–Detemple adjacent-averaged tree oracle, with the Volterra level-set solver demoted to a method under test and its grid sensitivity quantified; (vi) metric labels corrected (RMSE / true MAE / maximum absolute error). On a 90-configuration grid (450 prices, $T \leq 3$, $\sigma \leq 0.6$) the revised formula attains RMSE 1.5×10^{-3} , mean absolute error 8×10^{-4} , maximum error 7.8×10^{-3} against the oracle, versus 0.279/0.162/1.131 for Barone-Adesi–Whaley. No closed form for the true free boundary is claimed.

Table of contents

1	Revision record	2
2	Setup and exact machinery	2
3	Short-expiry boundary asymptotics: three regimes	3
4	The boundary family (redesigned)	4
5	Collocation: a reported projection, no stationarity	4
6	The Volterra solver and its known limits	4
7	Numerical results	5
8	Scope and limitations	6

1 Revision record

This is Revision 2 of the Phase 15 manuscript, responding point by point to `phase15-mathematical-review.md` (2026-09-02):

1. **Put sign.** Section 2.2 states the call and put premiums separately and notes the outer factor ϕ required by any unified one-line form.
2. **Short-expiry asymptotics.** Section 3 replaces the universal claim with the three-regime results of Evans et al. (2001) and Chen and Zhu (2010), and shows our numerics reproduce them.
3. **Boundary family.** Section 4 removes the $|\ln \max(\tau, \varepsilon)|$ term (artificial cusp at $\tau = 1$, dimensionful logarithm, nonmonotone candidates) and replaces it with smooth dimensionless regime-aware bases.
4. **Stationarity.** All envelope/stationarity claims are deleted; Section 5 demonstrates first-order sensitivity of the substituted-boundary functional.
5. **Reference solver.** Section 6 documents the solver's flat-step fallbacks and grid sensitivity, and Section 7 benchmarks against a converged tree oracle instead.
6. **Metric labels.** RMSE / true MAE / MaxAE throughout; the worst configurations are identified explicitly.

2 Setup and exact machinery

2.1 Risk-neutral dynamics and the complex-GBM picture

Under the risk-neutral measure, $dS_t/S_t = (r - q)dt + \sigma dW_t$, and the Phase 4 one-driver complex process (Complex-Itô Random-Walk Research Project 2026)

$$\mathcal{Z}_t = \mathcal{Z}_0 \exp\left(\left[(r - q - \tfrac{1}{2}\sigma^2) + i\omega\right]t + (\sigma + i\beta)W_t\right) \quad (1)$$

has modulus $|\mathcal{Z}_t| = S_t$ (with $|\mathcal{Z}_0| = S_0$) and unwrapped phase $\Theta_t = \Theta_0 + \omega t + \beta W_t$. In the drift-matched gauge the phase is a deterministic affine function of log-price, $\Theta_t = \Theta_0 + (\beta/\sigma) \ln(S_t/S_0)$, so the exercise boundary becomes a moving phase barrier. **Scope, stated plainly:** this is an invertible reparameterization (for $\beta/\sigma > 0$; the exercise inequality reverses for $\beta < 0$, and at $\beta = 0$ phase carries no spot information). It does not reduce the optimal-stopping problem or add pricing information by itself; the substance of this paper is the boundary family and the EEP machinery that follow.

2.2 The early-exercise premium, with correct put sign

Let $\tau = T - t$ and $b(\tau)$ the boundary in time to maturity. With $d_{1,2}(S, B, s) = [\ln(S/B) + (r - q \pm \frac{1}{2}\sigma^2)s]/(\sigma\sqrt{s})$, the call and put premiums are

$$\text{EEP}^c(S, \tau) = \int_0^\tau \left[qSe^{-qs} \Phi(d_1(S, b(\tau - s), s)) - rKe^{-rs} \Phi(d_2(S, b(\tau - s), s)) \right] ds, \quad (2)$$

$$\text{EEP}^p(S, \tau) = \int_0^\tau \left[rKe^{-rs} \Phi(-d_2(S, b(\tau - s), s)) - qSe^{-qs} \Phi(-d_1(S, b(\tau - s), s)) \right] ds. \quad (3)$$

Equivalently, the unified form carries an **outer factor** ϕ for puts:

$$V_A = V_E + \phi \int_0^\tau \left[q S e^{-qs} \Phi(\phi d_1) - r K e^{-rs} \Phi(\phi d_2) \right] ds, \quad \phi = \begin{cases} +1 & \text{call} \\ -1 & \text{put.} \end{cases} \quad (4)$$

Omitting the factor gives a put *below* its European price (base case: 7.493 versus the correct 7.661 at $r = q = 0.05$, $\sigma = 0.2$, $T = 1$), an arbitrage violation; the implementation and a unit test enforce the correct sign. The value-matching boundary equation inherits the same factor.

Anchors (Kim 1990; Jacka 1991): $b_c(0^+) = K \max(1, r/q)$ (infinite for $q = 0$), $b_p(0^+) = K \min(1, r/q)$; perpetual boundaries $b_\infty^c = K s_+/(s_+ - 1)$, $b_\infty^p = K s_-/(s_- - 1)$ from the roots of $\frac{1}{2}\sigma^2 s(s-1) + (r-q)s - r = 0$ ($s_+ > 1$ when $q > 0$; at $q = 0$, $s_+ = 1$ and the call boundary is infinite, i.e. never exercise, recovering Merton (1973)).

3 Short-expiry boundary asymptotics: three regimes

The established results for the put boundary near expiry (Evans et al. 2001; Chen and Zhu 2010, eqs. (3.15)–(3.22)) are

$$\begin{aligned} q < r: \quad K - b_p(\tau) &\sim K \sigma \sqrt{\tau |\ln \tau|}, \\ q = r: \quad K - b_p(\tau) &\sim K \sigma \sqrt{2\tau |\ln \tau|}, \\ q > r: \quad b_p(\tau) &= \frac{r}{q} K \left[1 - \xi_1 \sigma \sqrt{2\tau} + O(\tau) \right], \quad \xi_1 \approx 0.4517, \end{aligned} \quad (5)$$

with calls inheriting mirrored regimes under $r \leftrightarrow q$. Fitting our solved near-expiry boundaries (flat fallback steps excluded and counted):

Table 1: Near-expiry regime fits from the Volterra solutions.

Regime	Fit ($\sqrt{2\tau \ln \tau }$ normalization)	Prediction	$\sqrt{2\tau}$ slope	Prediction
put, $r =$ 0.07, $q =$ 0.03 ($q < r$)	0.724	$1/\sqrt{2} =$ 0.707	—	—
put, $r = q =$ 0.05	1.044	1	—	—
put, $r =$ 0.03, $q =$ 0.07 ($q > r$)	0.09 (window artifact)	fits $\sqrt{\tau}$, not $\sqrt{\tau \ln \tau }$	0.0393	$(r/q)\xi_1\sigma =$ 0.0387

The earlier Phase 15 interpretation (“universal shape with $r - q$ -dependent coefficient”) is withdrawn: the measured values are precisely the known regimes, including the fact that fitting a $\sqrt{\tau |\ln \tau|}$ regressor to the $q > r$ square-root law produces an apparent coefficient that tends to zero as $\tau \downarrow 0$.

4 The boundary family (redesigned)

With $s_\phi = +1$ for calls and -1 for puts, write the signed coefficients $\eta_{\log} = s_\phi h_{\log}$, $\eta_{\text{root}} = s_\phi h_{\text{root}}$ with $h_{\log}, h_{\text{root}} \geq 0$:

$$\ln b(\tau) = \ln b_\infty + \ln\left(\frac{b_0}{b_\infty}\right) e^{-\kappa_1 \tau} + \sigma e^{-\kappa_2 \tau} \left[\eta_{\log} \psi_{\log}(\tau; \hat{\tau}) + \eta_{\text{root}} \psi_{\text{root}}(\tau) \right] \quad (6)$$

$$\psi_{\log}(\tau; \hat{\tau}) = \sqrt{2\tau \ln\left(1 + \frac{\hat{\tau}}{\tau}\right)}, \quad \psi_{\text{root}}(\tau) = \sqrt{2\tau}. \quad (7)$$

Properties, each enforced by a unit test:

- **Dimensionless.** $\hat{\tau}$ is a declared time scale in the same unit as τ ; changing units rescales both, and the price is unit-invariant (Section 7.2).
- **Smooth.** $\psi_{\log}$ is smooth on $(0, \infty)$; $\psi_{\log} \sim \sqrt{2\tau |\ln(\tau/\hat{\tau})|}$ as $\tau \downarrow 0$. The former term $|\ln \max(\tau, \varepsilon)|$ had an infinite-slope cusp at $\tau = 1$ — an arbitrary, dimension-dependent point — which is gone.
- **Regime-aware.** The two bases carry the logarithmic and square-root short-expiry shapes of Eq. 5; collocation allocates their weights.
- **Anchored.** $b(0^+) = b_0$ and $b(\tau) \rightarrow b_\infty$ for all parameter values.
- **Monotone in practice.** The old cusp produced $b(0.9) = 132.3 > b(1.0) = 129.9 < b(1.1) = 136.2$ for the $T = 3$, $r = q = 0.05$, $\sigma = 0.2$ call; the redesigned family gives a monotone boundary through that neighborhood (regression-tested). Monotonicity is a numerically verified property of the collocated members, not an analytic guarantee of the whole family.

5 Collocation: a reported projection, no stationarity

$(\kappa_1, \kappa_2, h_{\log}, h_{\text{root}})$ **minimize** the sum of squared value-matching residuals at four fractional nodes ($\tau \in \{0.2, 0.4, 0.6, 0.8\} \cdot T$) under box constraints, multi-start L-BFGS-B. This is a projection: residuals are generally *not* zero, and the published record reports, per calibration, the objective, the four residuals, the optimizer status, and the multi-start sensitivity. For the base case the residuals are $\sim 10^{-5}$ – 10^{-4} ; degenerate long-maturity cases can leave residuals at the 10^{-2} level, which is reported, not hidden.

No stationarity claim. The substituted-boundary functional $V_E + \text{EEP}[b]$ is first-order sensitive to boundary error: for $d_{1,2}$ as above, the identity $Se^{-qs}\varphi(d_1) = Be^{-rs}\varphi(d_2)$ gives $\partial \text{integrand} / \partial B = e^{-rs}\varphi(d_2)(rK - qB)/(B\sigma\sqrt{s})$, so the first variation under a perturbation h is generally nonzero. Numerically, with $b_\varepsilon(u) = b(u)e^{\varepsilon \sin(\pi u/T)}$ on the base call, $(V[b_\varepsilon] - V[b])/\varepsilon$ converges to ≈ -0.345 as $\varepsilon \rightarrow 0$ through $10^{-2}, 5 \times 10^{-3}, 2.5 \times 10^{-3}, 1.25 \times 10^{-3}$ — first order, not second. Accuracy below is established by direct benchmarking against the oracle, not by any envelope argument.

6 The Volterra solver and its known limits

The causal sequential level-set solver (Phase 15, revised root logic) locates, at each grid point, where the value-matching residual reaches the level $-\delta K$ on the continuation side — the exact residual is identically zero on the exercise side ($F \equiv 0$ for $y \geq b$ for calls, $y \leq b$ for puts; the previously claimed put “positive hump” is a numerical artifact and is no longer used). Its limits, stated plainly and instrumented (SOLVER_DIAGNOSTICS):

- **Flat fallbacks.** Points without a bracketed level crossing copy the previous value; counts are reported. Near $\tau = 0$ the equation loses traction ($\partial F / \partial y \rightarrow 0$), and in long-maturity hard regimes the fallback fraction can exceed one half of all steps.

- **Monotonicity is enforced** by clipping toward the perpetual limit: monotone output is a solver property, not an independent check.
- **Grid sensitivity.** In the benchmark’s hard regimes the solver is not a reliable reference (Table 2):

Table 2: Grid behavior of the Volterra solver against the tree oracle. The $n = 300$ value is an isolated outlier of ≈ 0.30 ; fallback counts exceed 200 steps in this configuration.

Settings	Price ($S = 120, r = 0.05, q = 0.10, \sigma = 0.2, T = 3$, call)
Solver $n = 250$	20.634
Solver $n = 300$	20.392
Solver $n = 350$	20.677
Solver $n = 700$	20.671
Adjacent-averaged CRR	20.692 (spread 1.1×10^{-3})
$N \in \{8000, 10000, 12000\}$ (mean)	

Consequently **all benchmarks use the converged Broadie–Detemple adjacent-averaged tree** (Broadie and Detemple 1996) as the oracle ($N \in \{8000, 12000\}$, mean; spread reported per row), with the solver demoted to one method under test. (Prices also remain sensitive at the 10^{-2} level to environment-level floating-point differences, e.g. numpy SIMD paths — one more reason the tree, not the solver, is the oracle.)

7 Numerical results

7.1 Benchmarks against the oracle

Reduced grid: 18 configurations $\times$ 3 spots ($S \in \{90, 100, 110\}$). Extended grid: 90 configurations $\times$ 5 spots ($S \in \{80, \dots, 120\}$, $\sigma \in \{0.2, 0.4, 0.6\}$, $T \in \{0.5, 1, 3\}$, five (r, q) pairs, both kinds). Metrics against the oracle, labeled correctly:

Table 3: Benchmark metrics against the converged tree oracle.

Grid	Method	RMSE	MAE (mean)	MaxAE
Reduced	spiral formula	0.00066	0.00043	0.0028
Reduced	Barone-Adesi–Whaley	0.161	0.096	0.427
Reduced	Volterra solver ($n = 300$)	0.0077	0.0039	0.031
Extended	spiral formula	0.00147	0.00081	0.0078
Extended	Barone-Adesi–Whaley	0.279	0.162	1.131
Extended	Volterra solver ($n = 300$)	0.0206	0.0070	0.301

The worst spiral case on the extended grid is $(r, q, \sigma, T) = (0.05, 0.10, 0.2, 1)$ call at $S = 120$ (error -0.0078); the worst BAW errors concentrate in deep-ITM long-maturity puts, e.g. the $T = 3$, $\sigma = 0.2$, $r = 0.07$, $q = 0.03$ put at $S = 80$ identified by the review. Raw formula values are used throughout — no clipping — and deep-ITM raw values that sit slightly outside no-arbitrage bounds are visible in the CSVs as diagnostic information.

7.2 Structural identities

- **Time-unit invariance.** The same option priced in years and in months (rates per month, $\sigma/\sqrt{12}$, $\hat{\tau} = 12$ months) agrees to 4.5×10^{-5} .

- **Off-ATM put–call duality.** $P(90, 100; 0.07, 0.03)$ versus the swapped call $C(100, 90; 0.03, 0.07)$, each on its own solved boundary: agreement to 2.0×10^{-3} .
- **Zero-dividend call.** Returns the European price to machine precision.

8 Scope and limitations

- **No closed form for the true boundary is claimed.** The constant-coefficient free-boundary problem remains open; this paper delivers a collocation formula on an engineered family, validated against a converged oracle. The honest one-line description: a benchmark-validated collocation formula built on standard EEP machinery and an invertible complex-phase reparameterization of log-price.
- **Solver sensitivity.** The Volterra solver is grid-sensitive in hard regimes (Table 2) and carries a heavy flat-step fraction near maturity; it is not the oracle anywhere in this revision.
- **Raw formula values.** No clipping: approximate-boundary prices can sit slightly outside no-arbitrage bounds; the deviations are reported, and the upper bounds $C_A \leq S$, $P_A \leq K$ are not enforced by the formula.
- **Near-expiry fits.** The regime fits exclude flat fallback steps and are only as reliable as the remaining point count (reported alongside).
- **Model scope.** Constant r, q, σ , continuous dividends, standard single-asset American calls/puts.

9 Reproducibility

All computations run in the existing **phase3-paper** conda environment (Python 3.12, numpy, scipy, matplotlib); no packages installed. Fixed seed 20260902. Artifacts:

- `phase15_american.py` — EEP machinery, regime-aware spiral family, collocation, solver with diagnostics, BAW, tree oracle;
- `tests/test_phase15_american.py` — 22 tests including the review’s requested coverage (put sign, off-ATM duality, regime asymptotics, time-unit invariance, monotonicity through $\tau = 1$, no hidden clipping, grid behavior);
- `notebooks/build_phase15_american_spiral.py` → executed notebook — the computational record behind every number above (`phase15_figures/`, `phase15_tables/`, including the extended oracle CSV);
- `paper/phase15/` — this manuscript.

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